



Stony Brook University

Utility-based Portfolio Optimization

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INTRODUCTION

- **Portfolio Optimization:** Allocation of assets to balance risk and return.
- **Discrete Case:** Classical Markowitz mean-variance optimization formulated in discrete time.
- **Continuous Case:** Continuous-time portfolio optimization using stochastic control and HJB equations.
- **Utility-Based Portfolio Optimization:** Maximization of the expected utility of terminal wealth.
- **Analytic Solution:** Closed-form solutions for utility-based portfolio optimization.
- **Physics-Informed Neural Networks (PINN):** Using PINNs to solve HJB PDEs governing optimal investment strategies.
- **Monte Carlo and PINN Simulation Analysis:** Comparing numerical simulations and out-of-sample performance.
- **Sensitivity Analysis:** Optimal strategies and model outputs respond to changes in key parameters.



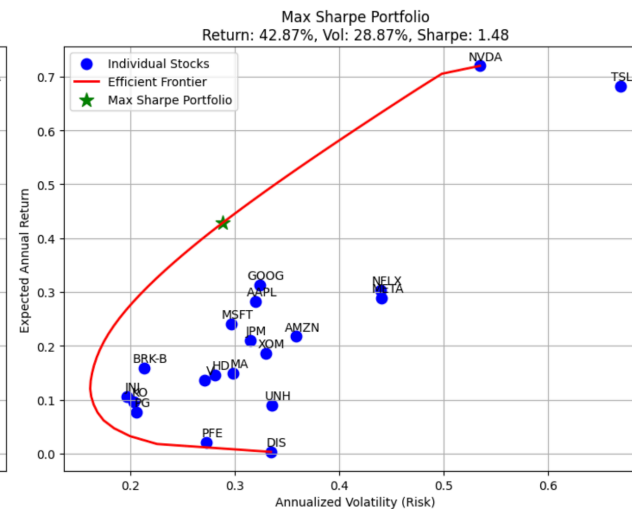
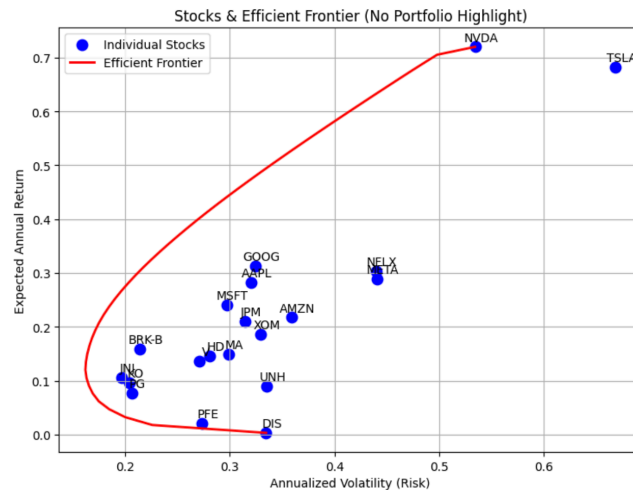
PORTFOLIO OPTIMIZATION

- Process of selecting the best combination of assets to maximize returns for a given level of risk.
- Balances the trade-off between expected return (μ) and risk (σ) to achieve efficient portfolios.
- Design strategies that improve long-term performance and reduce downside risk.
- Includes discrete-time Markowitz models, continuous-time stochastic control, and utility-based approaches.
- Modern techniques use PDEs, Monte Carlo, and deep learning (PINNs) to solve optimization problems.



DISCRETE PORTFOLIO OPTIMIZATION

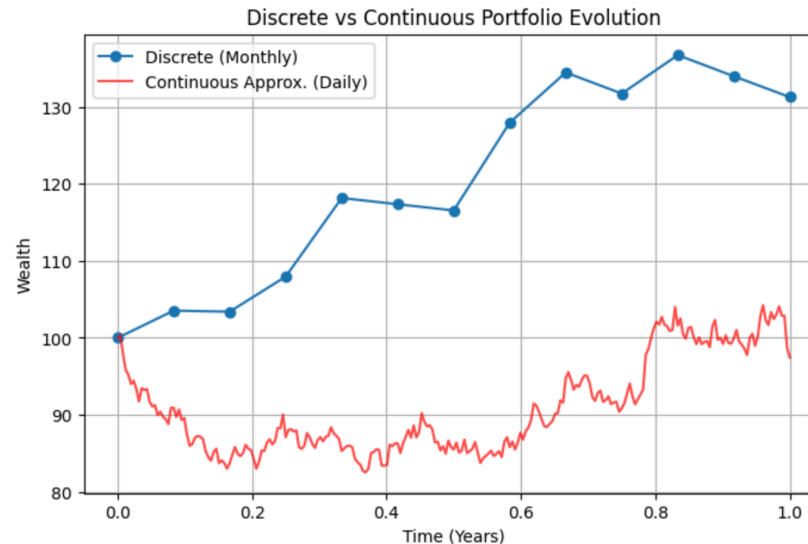
- Portfolio decisions are made at fixed time intervals, daily, monthly, or annually.
- Uses estimated returns and covariances to construct portfolios that optimize risk-return trade-offs.
- Typically solved with Markowitz mean–variance theory, scenario-based numerical methods etc.





CONTINUOUS-TIME PORTFOLIO OPTIMIZATION

- Portfolio allocation is adjusted continuously over time.
- Modelled using stochastic differential equations (SDEs).
- Optimal strategies are derived by solving Hamilton–Jacobi–Bellman (HJB) equations.
- Includes advanced features like utility-based objectives and constraints.





CONTINUOUS-TIME PORTFOLIO OPTIMIZATION

- Investor allocates wealth X_t between a risk-free asset B_t and a risky stock S_t
- Risk-free asset dynamics:

$$dB_t = rB_t dt, \quad \text{where, } r \text{ is the risk-free rate}$$

- Risky asset (stock) dynamics:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad \text{with drift } \mu, \text{ volatility } \sigma, \text{ and } W_t \text{ a standard Brownian motion.}$$

- Wealth dynamics under allocation π_t :

$$dX_t = rX_t dt + \pi_t(\mu - r)dt + \pi_t\sigma dW_t$$

- π_t dollar amount invested in the risky asset
- Remaining wealth $X_t - \pi_t$ goes to the risk-free asset
- $rX_t dt$, interest earned on total wealth
- $\pi_t(\mu - r)dt$, extra expected return from holding the risky asset
- $\pi_t\sigma dW_t$, random gain/loss from the risky asset



UTILITY FUNCTION

- Instead of just maximizing return or minimizing risk, a utility function helps find the portfolio that gives the highest satisfaction to that specific investor.

- Mean–Variance Utility (Quadratic Utility)

$$U = \mu - \frac{\gamma}{2}\sigma^2$$

- Exponential Utility
- Constant Absolute Risk Aversion (CARA)
- Closed-form solutions exists in continuous time

$$U(X_t) = -e^{-\gamma x}, \gamma > 0$$

- Power Utility
- Constant Relative Risk Aversion (CRRA)

$$U(X_t) = \frac{x^{1-\gamma}}{1-\gamma}$$

- γ = risk-aversion parameter



UTILITY FUNCTION

- Utility-based optimization focuses on investor preferences toward risk.
- Portfolio optimization becomes:

$$\sup_{\pi_t} \mathbb{E}[U(X_T)]$$

- chooses a strategy, π_t that maximizes expected satisfaction from terminal wealth.
- If $V(t, x)$ is value function,

$$V(t, x) = \sup_{\pi_t} \mathbb{E}[U(X_T) \mid X_t = x]$$

- Maximum achievable expected utility for given current wealth x and time t

$$V(t, x) = \sup_{\pi_t} \mathbb{E}[V(t + dt, X_{t+dt}) \mid X_t = x]$$

- Optimal decision now depends on maximizing the expected value of future wealth.



ITÔ'S LEMMA

- Wealth dynamics under allocation π_t :

$$dX_t = rX_t dt + \pi_t(\mu - r)dt + \pi_t\sigma dW_t$$

- Apply Itô's lemma to the value function $V(t, x)$

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 V}{\partial x^2} (dX_t)^2$$

- Substitute dX_t and simplify

$$dV = \left[\frac{\partial V}{\partial t} + rX_t \frac{\partial V}{\partial x} + \pi_t(\mu - r) \frac{\partial V}{\partial x} + \frac{1}{2} (\pi_t\sigma)^2 \frac{\partial^2 V}{\partial x^2} \right] dt + (\pi_t\sigma \frac{\partial V}{\partial x}) dW_t$$

- Maximize the expected change in $V(t, x)$

$$\frac{\partial V}{\partial t} + \sup_{\pi} \left\{ rx \frac{\partial V}{\partial x} + \pi(\mu - r) \frac{\partial V}{\partial x} + \frac{1}{2} (\pi\sigma)^2 \frac{\partial^2 V}{\partial x^2} \right\} = 0$$

- Terminal condition

$$V(t, x) = U(x)$$



HAMILTON-JACOBI-BELLMAN (HJB)

- HJB PDE is,

$$\frac{\partial V}{\partial t} + \sup_{\pi} \left\{ rx \frac{\partial V}{\partial x} + \pi(\mu - r) \frac{\partial V}{\partial x} + \frac{1}{2}(\pi\sigma)^2 \frac{\partial^2 V}{\partial x^2} \right\} = 0$$

Terminal condition, $V(T, x) = U(x)$

- $rx \frac{\partial V}{\partial x}$, drift from risk-free growth
- $\pi(\mu - r) \frac{\partial V}{\partial x}$, improvement from taking risk
- $\frac{1}{2}(\pi\sigma)^2 \frac{\partial^2 V}{\partial x^2}$, penalty from risk
- The supremum chooses the best portfolio allocation



OPTIMAL ALLOCATION

- HJB PDE is,

$$\frac{\partial V}{\partial t} + \sup_{\pi} \left\{ rxV_x + \pi(\mu - r)V_x + \frac{1}{2}(\pi\sigma)^2 V_{xx} \right\} = 0$$

- Derivative w.r.t. π

$$\begin{aligned} \frac{d}{d\pi} \left[\pi(\mu - r)V_x + \frac{1}{2}(\pi\sigma)^2 V_{xx} \right] &= 0 \\ (\mu - r)V_x + \pi\sigma^2 V_{xx} &= 0 \end{aligned}$$

- Solving for π

$$\pi^*(t, x) = - \frac{(\mu - r)V_x}{\sigma^2 V_{xx}}$$

- Risk-return trade-off, and shows how the optimal strategy depends on wealth and risk aversion



ANALYTICAL SOLUTION

- If $f(t)$ is a deterministic function of time

$$V(t, x) = -e^{-\gamma x} f(t)$$

- Replace, $V_x = -\gamma V$, $V_{xx} = \gamma^2 V$

$$\pi^* = -\frac{(\mu - r)V_x}{\sigma^2 V_{xx}} = \frac{\mu - r}{\gamma \sigma^2}$$

- Plug π^* , into the HJB equation :

$$f'(t) = A f(t), \quad A = \frac{(\mu - r)^2}{2\gamma \sigma^2}$$

- Solving ODE:

$$f(t) = \exp(A(T - t))$$

- Final value function:

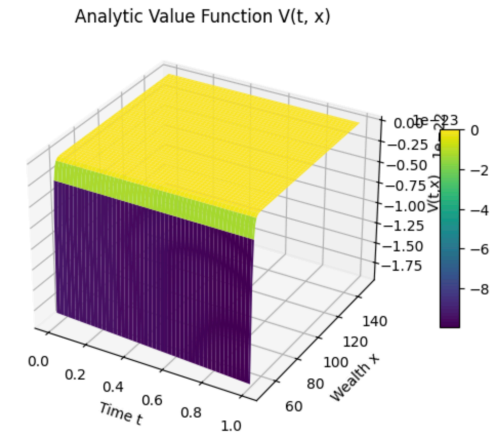
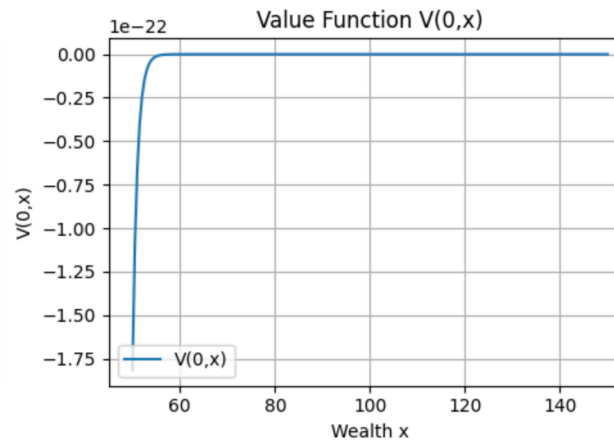
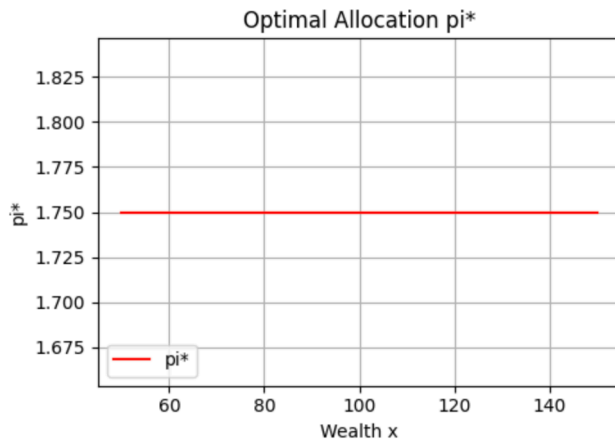
$$V(t, x) = -\exp\left[-\gamma x - \frac{(\mu - r)^2}{2\gamma \sigma^2}(T - t)\right]$$



ANALYTICAL RESULTS

- $\mu = 0.10$, $r = 0.03$, $\sigma = 0.20$, $\gamma = 1.0$, $T = 1.0$
- $\pi^*_{\text{analytic}} = 1.75$
- π^* , does NOT depend on wealth

CARA Analytic Solution: Optimal Allocation and Value Function





PHYSICS-INFORMED NEURAL NETWORKS

- Universal function approximators that can embed the knowledge of any physical laws
- Govern a given data-set in the learning process, which can be described by PDEs
- PDE is non-linear, may not have a closed-form solution in complex cases
- PINNs approximate the solution $V(t, x)$ with a neural network $V_\theta(t, x)$
- **PDE residual:** The network is trained so that the HJB equation holds at sampled points:

$$Residual(t, x) = \frac{\partial V}{\partial t} + \sup_{\pi} \left\{ rx \frac{\partial V}{\partial x} + \pi(\mu - r) \frac{\partial V}{\partial x} + \frac{1}{2}(\pi\sigma)^2 \frac{\partial^2 V}{\partial x^2} \right\} = 0$$

- **Boundary/Terminal condition:** The network satisfies

$$V(T, x) = U(x)$$

- **Optimization:** Minimize the combined loss:

$$Loss = MSE(PDEresidual) + MSE(boundarycondition)$$



PINN IMPLEMENTATION

- Two Neural Networks Approximate the Solution
- Interior points are sampled randomly
- Autograd computes V , V_x , and V_{xx}
- Terminal condition at $t = T$ included in total loss
- Boundary conditions are imposed at the boundary to stabilize training
- Additional regularizer drives toward analytic maximum



PINN IMPLEMENTATION

- $V(t, x)$: the value function of wealth at time t
- $\pi(t, x)$: the optimal portfolio allocation

$$\text{Loss} = w_{\text{PDE}} L_{\text{PDE}} + w_{\text{term}} L_{\text{terminal}} + w_{\text{bc}} L_{\text{boundary}} + w_{\pi} L_{\text{policy-reg}}$$

$$\bullet L_{\text{PDE}} = \mathbb{E} \left[\left(V_t + rxV_x + \pi(\mu - r)V_x + \frac{1}{2}(\pi\sigma)^2 V_{xx} \right)^2 \right]$$

$$\bullet L_{\text{terminal}} = \mathbb{E} \left[\left(V(T, x) - (-e^{-rx}) \right)^2 \right]$$

$$\bullet L_{\text{boundary}} = \mathbb{E} \left[\left(V(t, x_{\min}) - V_{\text{target}}(x_{\min}) \right)^2 + \left(V(t, x_{\max}) - V_{\text{target}}(x_{\max}) \right)^2 \right]$$

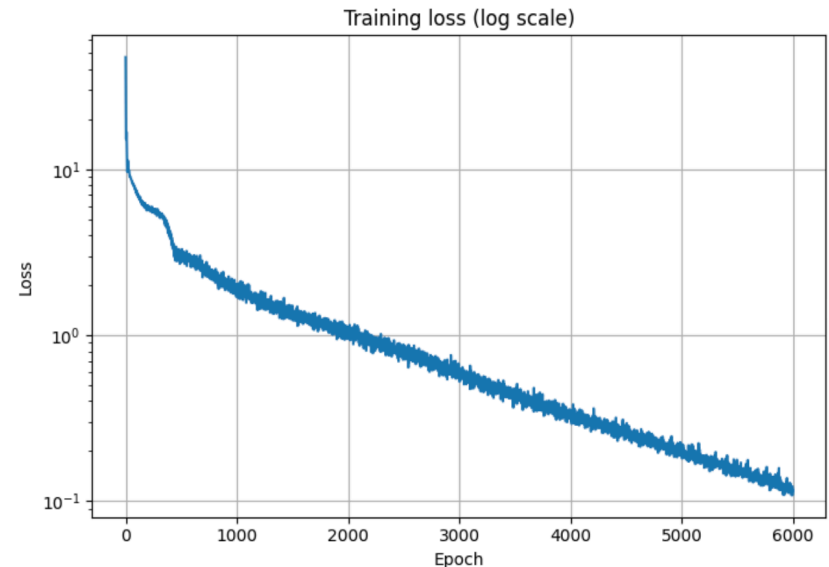
$$\bullet L_{\pi} = \mathbb{E} \left[\left(\pi(t, x) - \pi^* \right)^2 \right]$$



PINN IMPLEMENTATION

- Wealth, $x \in [x_{\min}, x_{\max}]$
- Number of collocation points, N_f
- Number of samples on the terminal boundary $t = T, N_T$
- Number of samples on the spatial boundaries, N_b
- Loss Function Weights, w_{PDE} , w_{terminal} , w_{boundary}

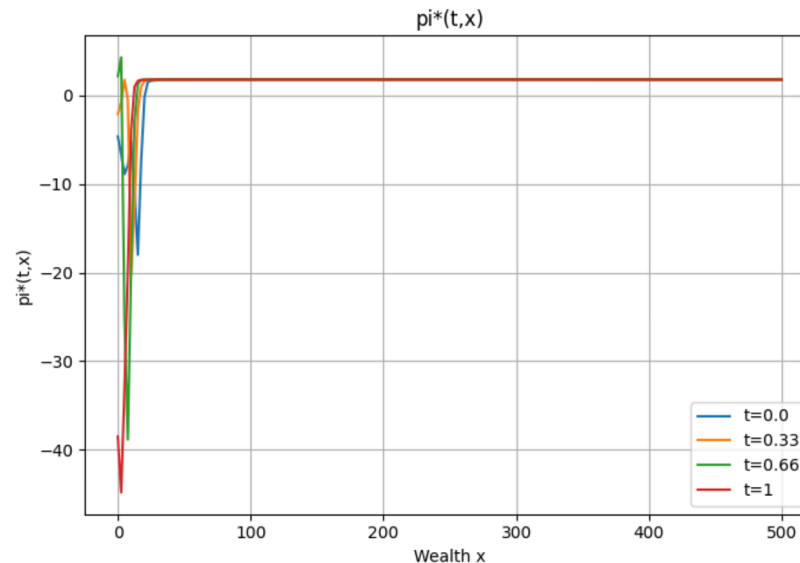
Parameter	Value
$x_{\min}$	0
$x_{\max}$	500
N_f (interior points)	5000
N_T (terminal points)	300
N_b (boundary points)	200
Learning rate (lr)	5×10^{-4}
Epochs	6000
Hidden layer size	64
Number of layers (nlayers)	3
Activation function	Tanh
w_{PDE}	1.0
w_{terminal}	200.0
w_{boundary}	10.0
w_{π} (policy regularizer)	10^{-4}





PINN RESULTS

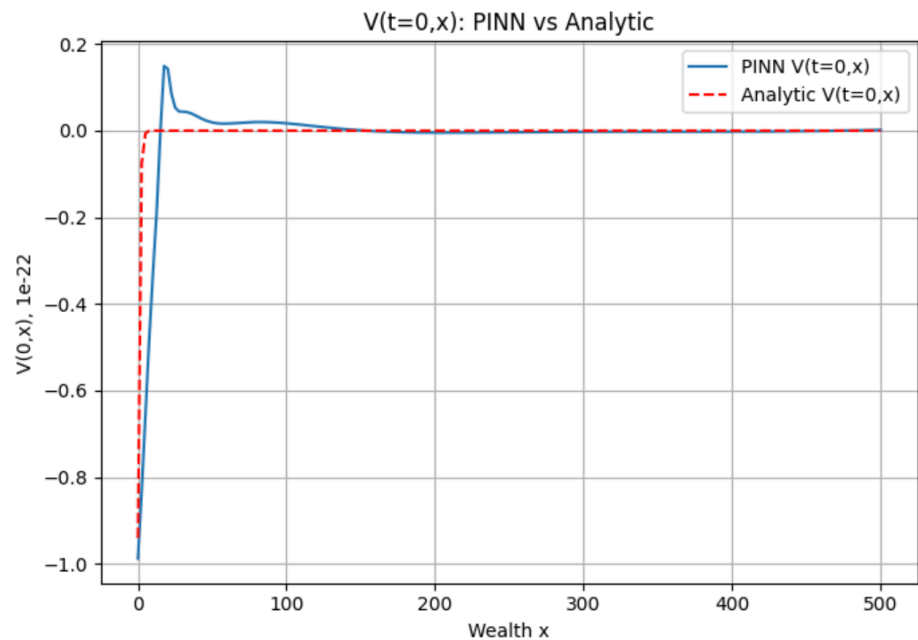
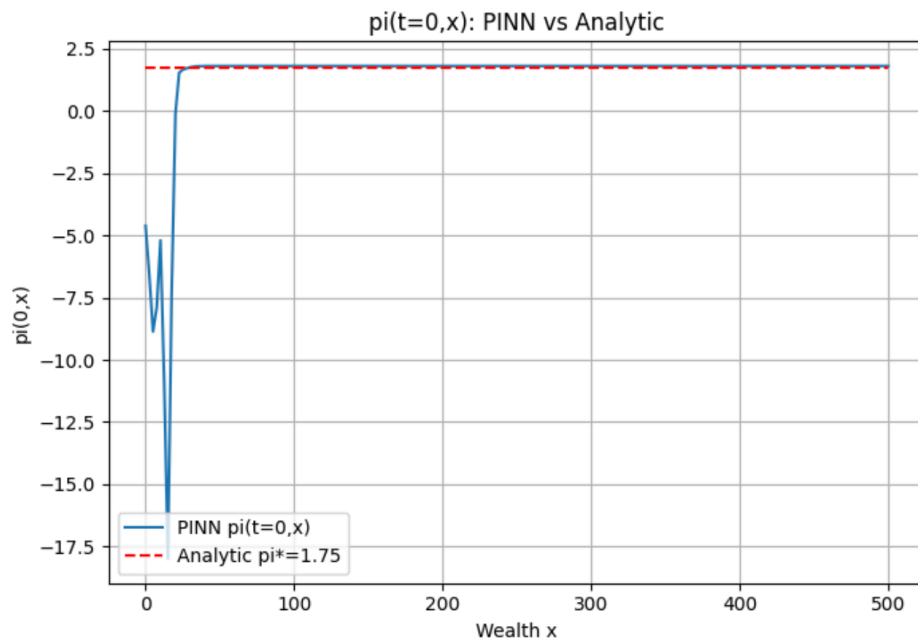
- Curves at different times reveal how the strategy evolves as time approaches maturity
- Nearly flat curves indicate the model is close to the analytic solution
- No fluctuations for $t=1$
- No deviations indicate the absence of approximation error and PDE satisfaction





PINN RESULTS

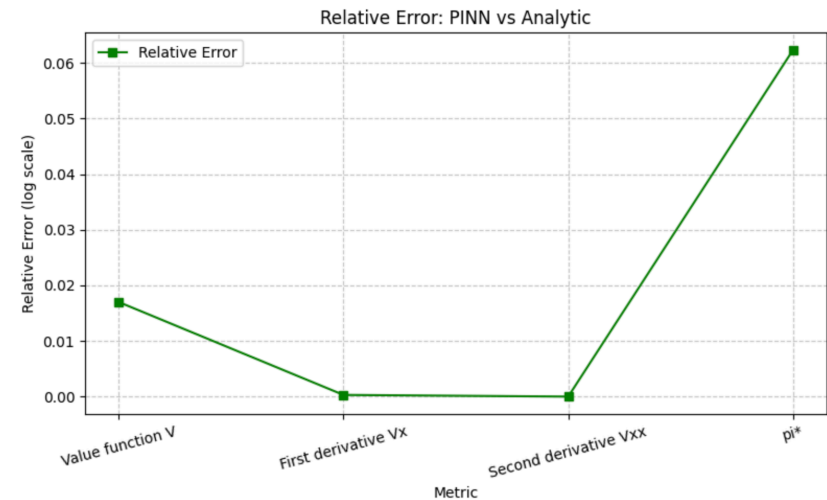
- $\pi(t=0, x)$ and $V(t=0, x)$ closely matches the analytical solutions across the wealth range.
- Small errors near the edges of the wealth domain, due to boundary effects or finite collocation points.





PINN RESULTS

- Analytic values of V , V_x , and V_{xx} are near zero
- Both V_x and V_{xx} show very small absolute errors
- The absolute and relative errors for π are $<1\%$

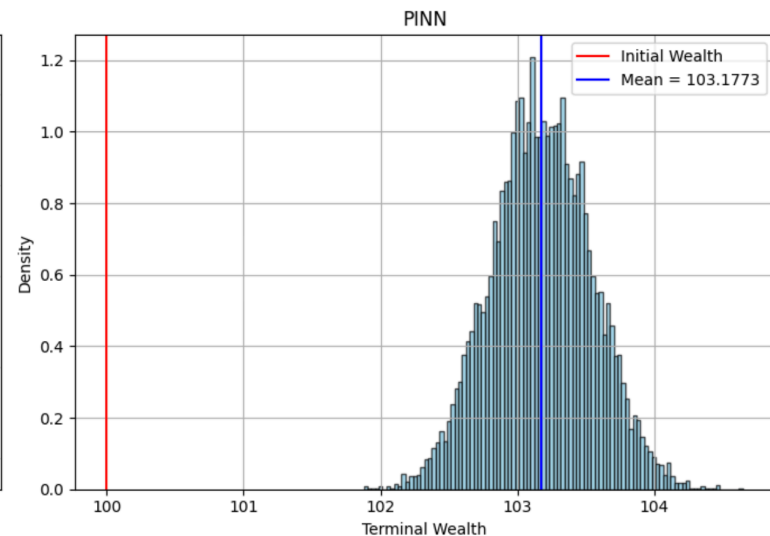
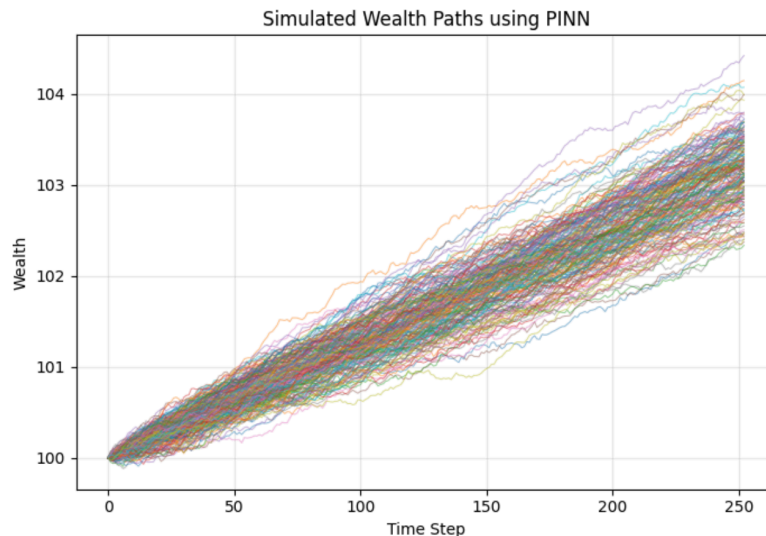


Metric	Analytic	PINN	Absolute Error	Relative Error
Value function V	-0.00	0.0271	0.0271	7.7344×10^{41}
First derivative V_x	0.00	0.0007	0.0007	2.1011×10^{40}
Second derivative V_{xx}	0.00	-0.0001	0.0001	2.3628×10^{39}
π^*	1.75	1.7616	0.0116	6.6×10^{-3}



MONTE CARLO WEALTH DYNAMICS

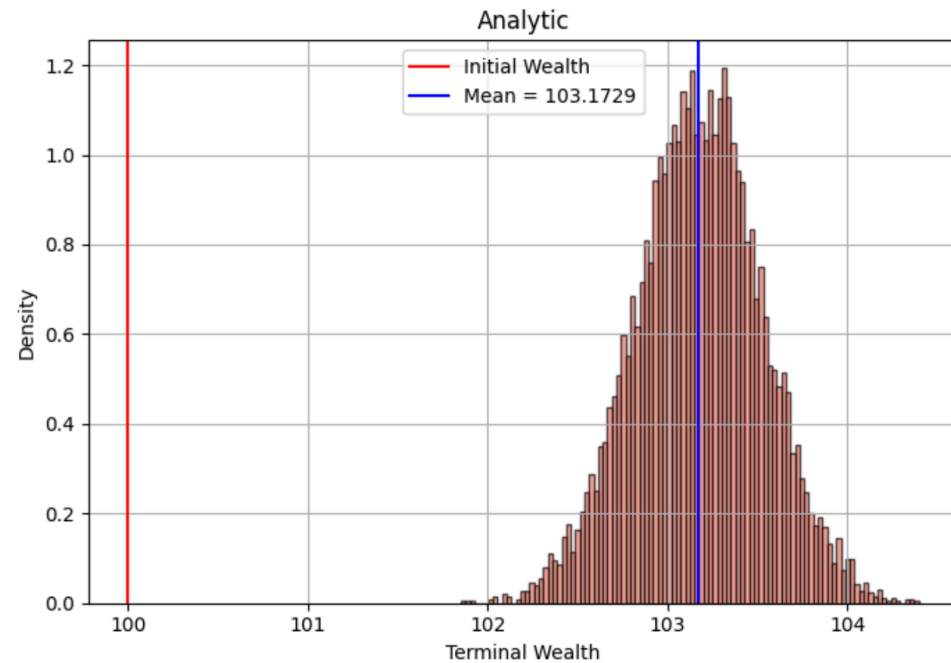
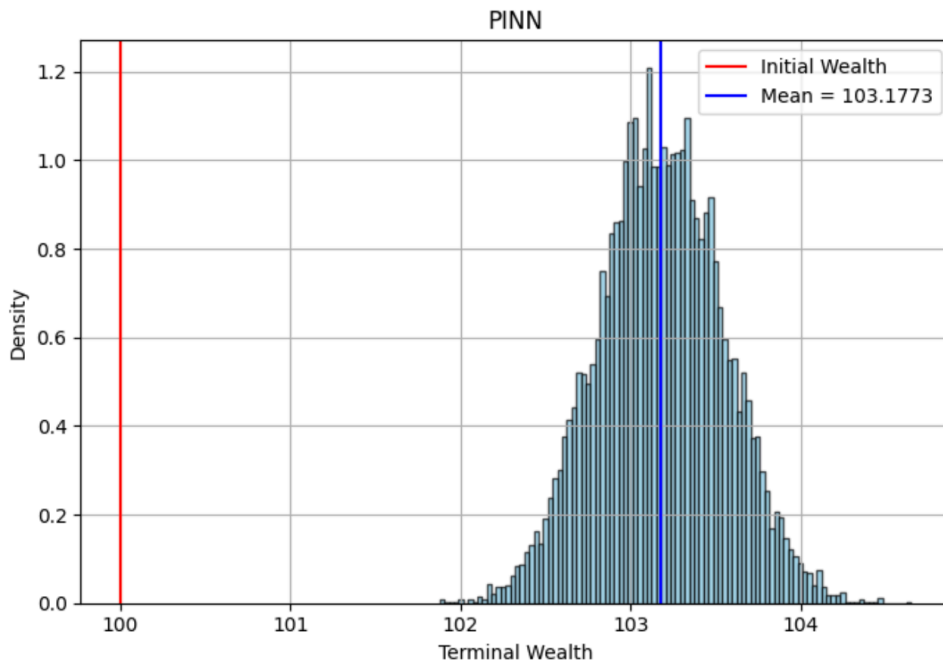
- **PINN input:** Takes current time t and wealth X_t as inputs
- **PINN output:** Predicts the optimal portfolio allocation $\pi(t, X_t)$
- **Wealth simulation:** Use $\pi(t, X_t)$ in Monte Carlo to update wealth X_t over discrete time steps.
- **Stochastic dynamics:** Random shocks dW_t are applied at each step to capture market uncertainty.





ANALYTICS VS PINN

- PINN Performance: Monte Carlo Simulation
- Analytic Performance: Closed-Form Merton Solution





ANALYTICS VS PINN

- PINN predicts an average terminal wealth nearly identical to the analytic value
- Standard deviation, expected utility, and certainty equivalent are very close to the analytic results
- Mean excess return and Sharpe ratio are comparable to analytic benchmarks

- GBM : $dX_t = (r + \pi(\mu - r))X_t dt + \pi\sigma X_t dW_t$
- $X_T = X_0 \exp \left[\left(r + \pi(\mu - r) - \frac{1}{2}(\pi\sigma)^2 \right) T + \pi\sigma W_T \right]$

Metric	Analytic Formula
Mean Terminal Wealth	$\mathbb{E}[X_T] = X_0 e^{rT} + \pi^*(\mu - r)T$
Std Dev Terminal Wealth	$\text{Std}[X_T] = \pi^* \sigma \sqrt{T}$
Expected Utility	$\mathbb{E}[U(X_T)] = \mathbb{E}[-e^{-\gamma X_T}]$
Certainty Equivalent (CE)	$\text{CE} = -\frac{1}{\gamma} \ln \mathbb{E}[U(X_T)]$
Mean Excess Return	$\frac{\mathbb{E}[X_T - X_0]}{T}$
Sharpe Ratio	$\text{SR} = \frac{\text{Mean Excess Return}}{\text{Std}[X_T - X_0]}$

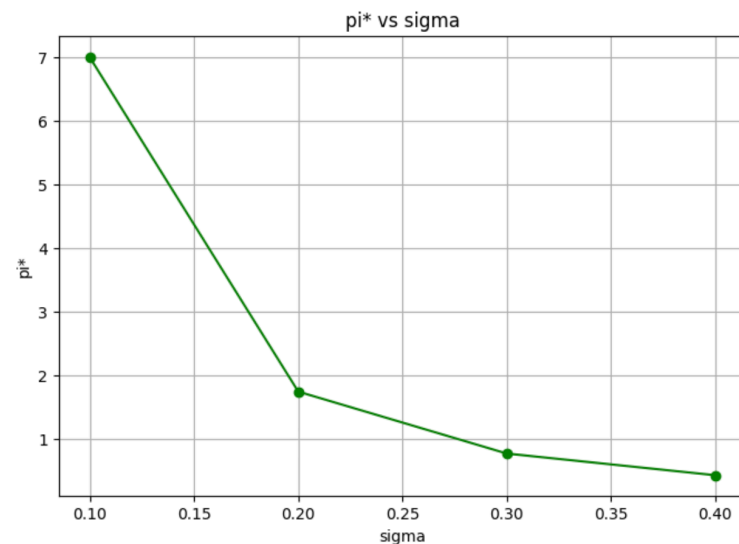
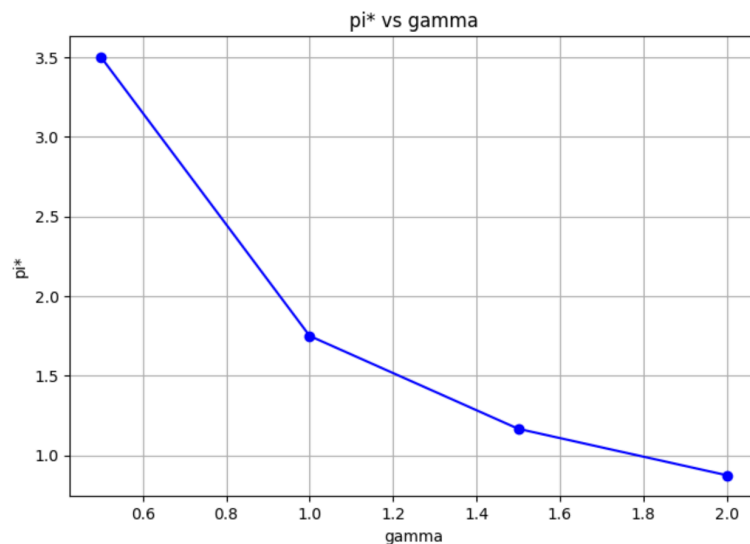
Metric	Analytic	PINN
Mean Terminal Wealth	103.16795	103.17120
Std Dev Terminal Wealth	0.35000	0.35552
Expected Utility	-0.00000	-0.00000
Certainty Equivalent (CE)	103.10670	103.10801
Mean Excess Return	3.16795	3.17120
Sharpe Ratio	9.05130	8.91993



SENSITIVITY ANALYSIS

- As γ increases, the optimal allocation π^* decreases, showing that investors invest less in the risky asset
- Higher σ leads to a lower π^* , showing reduced investment in the risky asset when market volatility rises

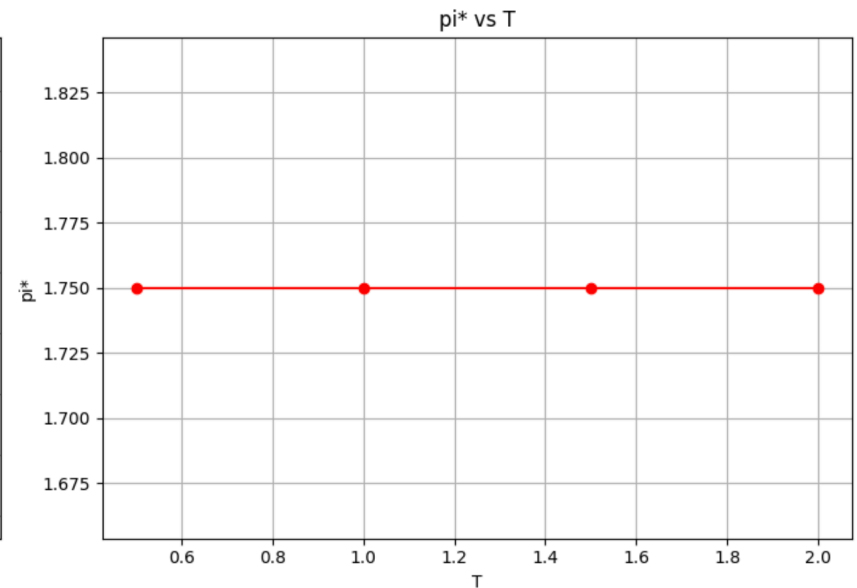
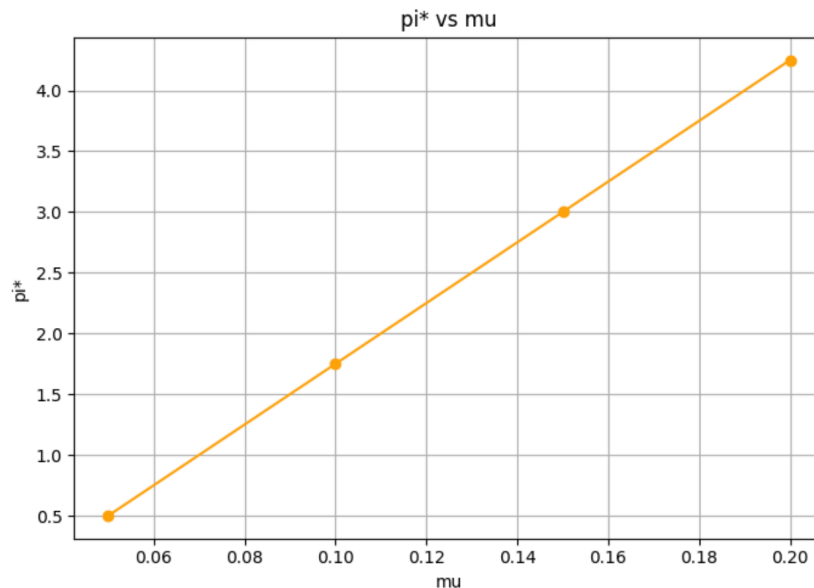
$$\pi^* = - \frac{(\mu - r)V_x}{\sigma^2 V_{xx}} = \frac{\mu - r}{\gamma \sigma^2}$$





SENSITIVITY ANALYSIS

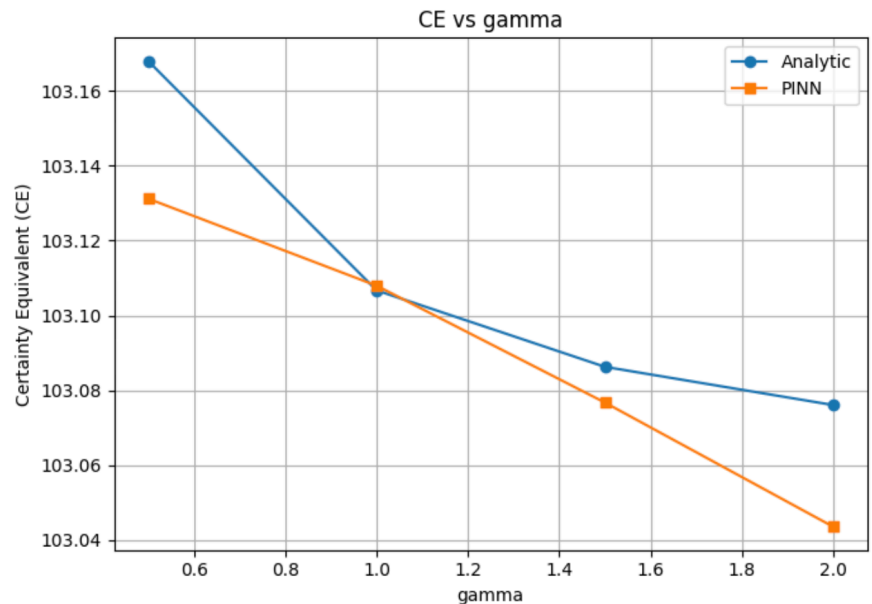
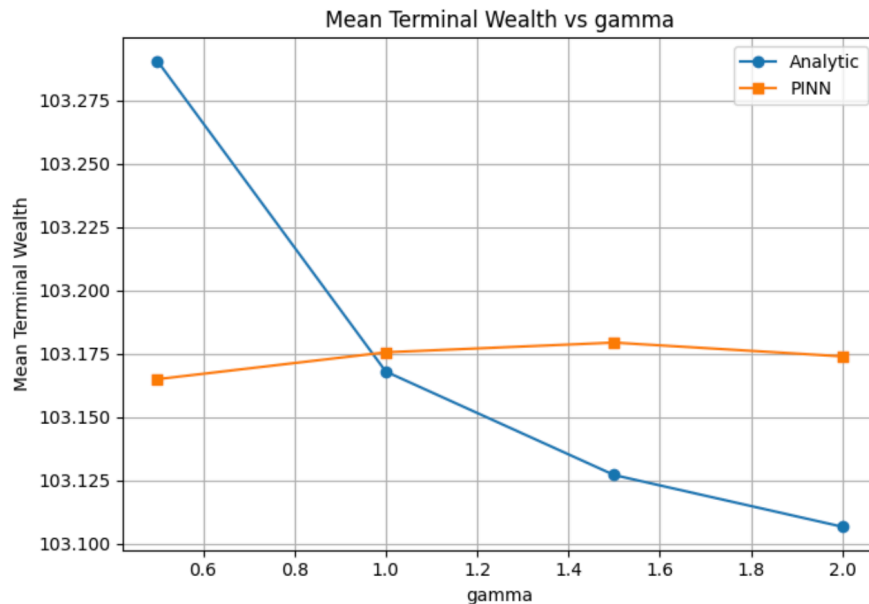
- π^* does not change with different investment horizons T .
- π^* increases linearly with μ
- Analytical π^* is sensitive to expected returns but unaffected by the length of the investment period





SENSITIVITY ANALYSIS

- The analytical mean terminal wealth decreases as γ increases.
- The PINN mean terminal wealth remains approximately stable with increasing γ
- The certainty equivalent decreases with larger γ in both the analytical and PINN results





CONCLUSION

- Shown π^* and $V(t,x)$ analytically
- π^* and $V(t,x)$ predicted by the PINN closely match the analytical solution
- The PDE residual and terminal conditions were not sufficient
- π regularization was applied.
- Performed sensitivity analysis on the analytical results and compared them with the PINN predictions



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